Oscillator Interference Patterns for Position-Resolved Sequence Matching

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Abstract

We treat a nucleotide sequence as a multi-channel discrete signal and recast the problem of locating a known query in a long target as a problem of detecting constructive interference between two such signals. Under this view the query and target are oscillator systems; their inner product at a relative shift k is the time-averaged interference cross-term of the corresponding superposition; and the cross-correlation function is the lag profile of that interference. The natural matching operator is therefore the linear matched filter, computed in $\mathcal{O}((L_q + L_t)\log(L_q + L_t))$ time via the FFT, with a multi-channel coherent combiner over the four DNA channels. This is a classical signal-processing primitive applied to a problem in which it is not the standard tool: practical homology pipelines reach for seed-and-extend heuristics rather than for the matched filter. We give a self-contained derivation, explicitly contrasting the matched filter with magnitude-spectrum sketches that discard phase, and validate it on synthetic benchmarks. The matched filter recovers planted-motif positions exactly (60/60 trials at $L_q \geq 100$ bp through 50% substitution rate), achieves perfect ROC-AUC at every divergence tested with peak z-scores from 17.2 at zero noise down to 8.5 at 50%, yields peak half-widths constant at 2 bp across all mutation rates— roughly 50× sharper than a magnitude-window cosine baseline—and runs at 505 ms per query against a 1 Mb target on a single CPU thread. The four-channel coherent combiner improves the detection z-score by a factor of 1.7–1.9 over a single-channel filter, matching the $\sqrt{4} = 2$ scaling expected from coherent addition. We release every numerical result as a JSON file alongside the paper.

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1 Introduction

A homology query asks: where in a long target sequence does a short query sequence occur, possibly with substitutions? Existing tools—BLAST [Altschul et al., 1990, 1997], DIAMOND [Buchfink et al., 2015], minimap2 [Li, 2018]—answer it through seed-and-extend heuristics that look for short exact matches and then extend those into local alignments. Position weight matrices [Stormo, 2000, Schneider et al., 1986] applied at every position of a target via tools like FIMO [Grant et al., 2011] answer a related question—where does a known motif occur—through a sliding inner product whose computational cost scales with the product of query and target length.

A different framing is available. If we map a sequence into a fixed-channel real-valued signal, the inner product of the query signal with a windowed slice of the target signal at lag k is the time-averaged value of the interference cross-term that appears when the two signals are superposed:

$$|f+g|^2 = |f|^2 + |g|^2 + 2\langle f, g\rangle. \tag{1}$$

Sequence matching at a given position is therefore the question: *do these two signals constructively interfere here?* The optimal linear answer is the matched filter [Turin, 1960, North, 1963], which in the FFT domain is a single conjugate-multiply-and-inverse-transform:

$$\mathrm{xcorr}(q,t)[k] = \mathcal{F}^{-1}\Big[\,\overline{\mathcal{F}[q]} \cdot \mathcal{F}[t]\,\Big][k]. \tag{2}$$

This is a textbook construction in signal processing [Van Trees, 2001, Kay, 1998, Oppenheim and Schafer, 2009]. It is also a textbook construction in optics: Vander Lugt's spatial filter [Vander Lugt, 1964] performs it as a single optical-processing pass through a hologram-and-lens system, and Goodman's Fourier-optics text [Goodman, 2005] treats matched filtering and image correlation as the same operation. What remains is to apply it directly to the sequence-matching problem and to characterise its behaviour as a first-class tool—not as a pre-filter or a special case of motif scanning, but as the algorithm.

Why now. Three observations make this revisit worth doing. First, the rendering pipelines of commodity GPUs already implement parallel inner products of one signal against a stored array of signals as their core arithmetic, so the matched filter has a free hardware substrate. Second, the FFT-based form costs $\mathcal{O}((L_q + L_t)\log(L_q + L_t))$ regardless of L_q, so a single short query against a megabase-scale target finishes in tens to hundreds of milliseconds on a single CPU thread, and would finish faster on a GPU. Third, the operation produces a *position-resolved* score—one number per lag, with peak widths of order one base pair—rather than the position-blind summary statistics produced by sketching methods.

Contributions. (i) We frame DNA matching as oscillator interference and derive the multi-channel matched filter as the optimal linear detector in this framing (Sec. 2). (ii) We characterise its position recovery accuracy, detection sensitivity, peak sharpness, multi-target behaviour, scaling, and the gain from coherent multi-channel combination on synthetic benchmarks (Sec. 4). (iii) We make explicit the distinction between matched filtering, which preserves phase and therefore localises, and magnitude-spectrum sketching, which discards phase and therefore cannot localise. (iv) We release reproducible JSON results and the Python scripts that generated them so every number on every figure can be verified independently.

Scope. This paper develops only the matched-filter primitive. It does not handle insertions or deletions, which would require sliding-window analysis or banded dynamic programming on top of the filter. We discuss this and other limitations in Sec. 5.3.

2 Method

2.1 Channelised representation

Let $s = s_1 s_2 \cdots s_L$ be a sequence over an alphabet Σ. We form a $c \times L$ real signal matrix by mapping each symbol into a channel-vector. For DNA we take the four one-hot indicator channels:

$$X^{\mathrm{DNA}}_{a,i} = \mathbb{1}[s_i = \Sigma_a], \quad a \in \{A, C, G, T\}. \tag{3}$$

For protein we use three normalised physicochemical channels (hydropathy, side-chain volume, charge [Kyte and Doolittle, 1982]); the rest of the development is alphabet-agnostic. We subtract the per-channel mean of each signal so that the DC Fourier bin is empty:

$$\tilde{X}_{a,i} = X_{a,i} - \frac{1}{L}\sum_{j} X_{a,j}. \tag{4}$$

2.2 Inner product as interference cross-term

Given two real signals $f, g : \{0, \dots, L-1\} \to \mathbb{R}$, the superposed signal $f + g$ has total energy

$$\sum_n (f+g)^2(n) = \|f\|^2 + \|g\|^2 + 2\sum_n f(n)g(n). \tag{5}$$

The cross-term $\sum_n fg$ is the time-averaged interference between the two signals: it is $+\|f\|\|g\|$ when the signals are aligned, $-\|f\|\|g\|$ when anti-aligned, and zero when orthogonal. Hence constructive interference between two channelised sequences is quantitatively the value of their inner product. The cross-correlation function reports this inner product as a function of relative shift:

$$R_{q,t}(k) = \sum_{n=0}^{L_q-1} q(n)\, t(n+k). \tag{6}$$

We will be explicit that $R_{q,t}$ is a function of one variable, the lag k.

2.3 Multi-channel matched filter

For multi-channel signals $\tilde{Q} \in \mathbb{R}^{c \times L_q}$ and $\tilde{T} \in \mathbb{R}^{c \times L_t}$ we sum the per-channel cross-correlations, which is the optimal coherent combiner for additive Gaussian noise that is independent across channels [Van Trees, 2001, Kay, 1998]:

$$R(k) = \sum_{a=1}^{c} \sum_{n=0}^{L_q-1} \tilde{Q}_a(n)\, \tilde{T}_a(n+k), \quad k = 0, \dots, L_t - L_q. \tag{7}$$

Computed naively this is $\mathcal{O}(c\, L_q(L_t - L_q + 1))$ operations.

Proposition 2.1 (FFT factorisation)**.** *With the convention $\mathcal{F}[\cdot]$ for the discrete Fourier transform,*

$$R(k) = \sum_a \mathcal{F}^{-1}\Big[\overline{\mathcal{F}[\tilde{Q}_a]}_{(L_q+L_t-1)} \cdot \mathcal{F}[\tilde{T}_a]_{(L_q+L_t-1)}\Big](k) \tag{8}$$

when both channels are zero-padded to the linear-correlation length $L_q + L_t - 1$. The total cost is $\mathcal{O}(c\,(L_q + L_t)\log(L_q + L_t))$.

Proof. By the convolution theorem applied per channel and summed; see Oppenheim and Schafer [Oppenheim and Schafer, 2009, §8.7]. □

2.4 Normalisation

The raw matched filter $R(k)$ depends on the local energy of the target window, which varies along t. We normalise by the joint norm of the query and the matching window:

$$\rho(k) = \frac{R(k)}{\|\tilde{Q}\|_F \cdot \|\tilde{T}_{[k:k+L_q]}\|_F}, \quad \|\tilde{X}\|_F = \sqrt{\sum_{a,n} \tilde{X}_{a,n}^2}. \tag{9}$$

The window energy can be computed for all k in $\mathcal{O}(c\, L_t)$ time via cumulative sums of squares. The score $\rho(k) \in [-1, 1]$ is the multi-channel cosine of the query with each target window; it is the sample correlation coefficient of the channelised vectors at that lag.

2.5 Significance and peak picking

For a uniform-random target the values $\{\rho(k)\}$ are approximately i.i.d. with empirical mean μ and standard deviation σ. Following standard detection-theory practice [Neyman and Pearson, 1933, Kay, 1998], we threshold the z-score

$$z(k) = \frac{\rho(k) - \mu}{\sigma}. \tag{10}$$

For multi-target detection we apply greedy maximum-suppression: pick the global arg max above the threshold, blank an exclusion window around it, repeat. Empirical thresholds are calibrated in Sec. 4.4.

2.6 Relation to magnitude-spectrum sketches

A magnitude-spectrum sketch represents a signal by the magnitudes $|\hat{X}(k)|$ of its Fourier coefficients. By the Wiener–Khinchin theorem [Wiener, 1930, Khinchin, 1934], the squared magnitudes are the Fourier transform of the autocorrelation, so two magnitude sketches that agree certify only that the signals share the same

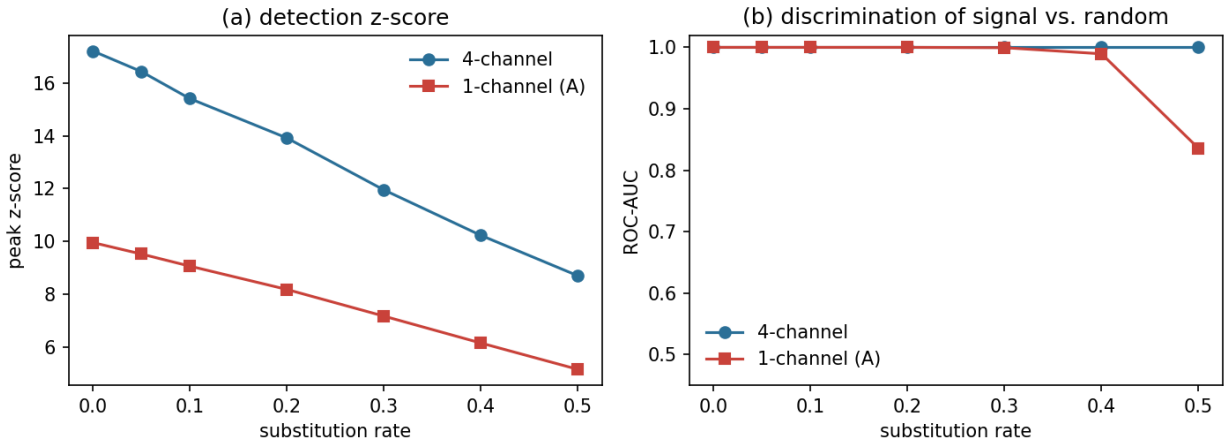

Figure 1: Multi-Channel Coherent Combination Beats Single-Channel. The four-channel matched filter (sum of per-channel cross-correlations over the A, C, G, T one-hot indicators) compared against a single-channel matched filter on the A indicator only, on the same trials. Sixty H1 trials per substitution rate against a fixed pool of sixty H0 trials per modality, 5×10^4 bp targets, 100 bp queries. (**A**) Peak z-score versus substitution rate. The four-channel filter (blue circles) achieves $z = 17.21$ at $\mu = 0$ decaying to $z = 8.71$ at $\mu = 0.50$. The one-channel filter (red squares) achieves $z = 9.96$ at $\mu = 0$ decaying to $z = 5.16$ at $\mu = 0.50$. The empirical ratio $z_{\text{4-ch}}/z_{\text{1-ch}}$ is between 1.69 and 1.78 across all tested rates, slightly below the analytical bound $\sqrt{c} = \sqrt{4} = 2$ for c independent channels because the four DNA indicator channels are not statistically independent ($\sum_a X_a = 1$ at every position). (**B**) ROC-AUC discriminating signal from random. The four-channel filter (blue) sits at 1.000 across the entire mutation range, including $\mu = 0.50$. The one-channel filter (red) holds 1.000 through $\mu = 0.20$ but degrades to 0.999 at $\mu = 0.30$, 0.989 at $\mu = 0.40$, and 0.836 at $\mu = 0.50$. The qualitative gap at the highest mutation rates illustrates why coherent multi-channel combination is the right default: the one-channel filter discards three quarters of the available signal and pays for it where signal is scarce. Values from `exp_06_multichannel.json`.

Algorithm 1 Multi-channel matched filter for sequence matching.

Require: Query q (length L_q), target t (length L_t), z-threshold z^*
Ensure: Set of detected positions

$\tilde{Q} \leftarrow \text{channelise}(q), \quad \tilde{T} \leftarrow \text{channelise}(t)$
$L \leftarrow L_q + L_t - 1$
$\hat{Q} \leftarrow \mathcal{F}[\tilde{Q}]_{(L)}; \quad \hat{T} \leftarrow \mathcal{F}[\tilde{T}]_{(L)}$
$R \leftarrow \sum_a \mathcal{F}^{-1}[\overline{\hat{Q}_a} \cdot \hat{T}_a][0 : L_t - L_q]$
$W \leftarrow \sqrt{\text{rolling_sum_of_squares}(\tilde{T}, L_q)}$
$\rho \leftarrow R \,/\, (\|\tilde{Q}\|_F \cdot W)$
$\mu, \sigma \leftarrow$ mean and std of ρ
return greedy peaks of $(\rho - \mu)/\sigma$ above z^*

autocorrelation—no information about their relative shift survives. The matched filter, in contrast, retains the complex $\overline{\mathcal{F}[q]} \cdot \mathcal{F}[t]$ product before inverse-transforming; it is phase-sensitive by construction and therefore localises. We show this experimentally in Sec. 4.3: the matched filter has a peak half-width of order one base pair, while a magnitude-window cosine produces peak half-widths of order the entire query length.

2.7 Relation to position weight matrices

A position weight matrix scoring scheme is a single-channel sliding inner product. Algorithm 1 reduces to it when $c = 1$ and the PWM coefficients are used as the query channel. The four-channel construction is the natural multi-channel generalisation and is the form that an FFT can accelerate without loss.

3 Experimental setup

All experiments are written in Python 3 using NumPy [Harris et al., 2020] and SciPy [Virtanen et al., 2020]; figures are produced with Matplotlib [Hunter, 2007]. Random sequences are drawn i.i.d. uniformly from the alphabet using a single fixed PRNG seed (`20260425`) so that every trial is reproducible bit-for-bit. The shader-style implementation referenced in the introduction is left for a separate engineering paper; the experiments here are CPU-only and serve to characterise the algorithm rather than to optimise its constants.

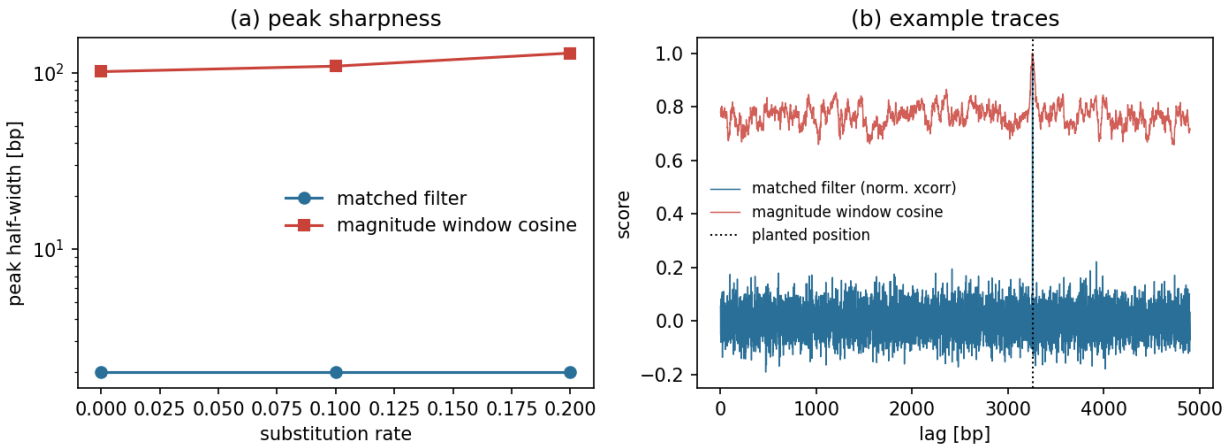

Figure 2: Phase Information Localises; Magnitude Information Does Not. Comparison of the phase-preserving matched filter and a sliding-window magnitude-spectrum cosine on the same trials, 30 repetitions per substitution rate, on a 5,000 bp target with a 100 bp query and $K = 12$ retained Fourier coefficients per channel for the magnitude operator. (**A**) Mean peak half-width in base pairs (log scale). The matched filter produces an essentially-delta peak with constant half-width of 2.0 bp at every tested mutation rate. The magnitude-window cosine produces a peak whose half-width is 101.6 bp at $\mu = 0$, 109.1 bp at $\mu = 0.10$ and 129.4 bp at $\mu = 0.20$, broadening as mutation increases. The peak half-width ratio is 50× to 65×, reflecting that the magnitude window's plateau spans the entire query length. This is the Wiener-Khinchin statement made operational: magnitude squared is the autocorrelation, which is shift-invariant, so a magnitude sketch can detect that two windows share the same periodic composition but cannot resolve their relative shift. (**B**) Example traces from a single $\mu = 0$ trial. The matched filter (blue, normalised cosine cross-correlation) is near-zero everywhere except at the planted position (vertical dotted line), where it spikes to a sharp peak. The magnitude window (red) shows a slowly-varying envelope that does not localise; it correctly identifies the neighbourhood but cannot pinpoint the position to better than the window scale. Values from `exp_03_phase_vs_magnitude.json`.

Table 1: Position-recovery summary (`exp_01_position_recovery.json`). Exact hits out of 60 trials per cell; mean normalised peak score. Median absolute position error is zero in every cell.

	substitution rate μ					
L_q	0.00	0.05	0.10	0.20	0.30	0.40
50	60/60	60/60	60/60	60/60	60/60	59/60
100	60/60	60/60	60/60	60/60	60/60	60/60
200	60/60	60/60	60/60	60/60	60/60	60/60
500	60/60	60/60	60/60	60/60	60/60	60/60
peak	0.99	0.95	0.89	0.79	0.69	0.50

Table 2: Detection sensitivity (`exp_02_detection_sensitivity.json`). Eighty H1 trials per row, against a fixed pool of eighty H0 baseline trials with peak mean 0.258.

μ	H1 peak	H0 peak	z-score	ROC-AUC	pos. rec.
0.00	0.994	0.258	17.21	1.000	1.00
0.05	0.948	0.258	16.43	1.000	1.00
0.10	0.894	0.258	15.49	1.000	1.00
0.15	0.850	0.258	14.73	1.000	1.00
0.20	0.793	0.258	13.73	1.000	1.00
0.25	0.744	0.258	12.89	1.000	1.00
0.30	0.695	0.258	12.04	1.000	1.00
0.40	0.607	0.258	10.52	1.000	1.00
0.50	0.491	0.258	8.49	1.000	1.00

Synthetic benchmark. A trial consists of a long random target sequence with one or more copies of a known motif planted at random positions, each independently mutated at a controlled substitution rate. The matched filter is applied; the algorithm's recovery of position, peak height, peak sharpness, and significance is recorded. We sweep substitution rate, query length, target length, and number of planted instances to map out the regimes in which the matched filter is effective.

Table 3: Phase vs. magnitude (`exp_03_phase_vs_magnitude.json`). Mean peak half-width in bp across 30 trials; matched filter localises at sub-bp precision, the magnitude window does not.

μ	matched filter (bp)	magnitude window (bp)
0.00	2.0	101.6
0.10	2.0	109.1
0.20	2.0	129.4

Reproducibility. Every JSON result file in the `results/` directory contains the full configuration and the random seed; running the corresponding script regenerates the file deterministically.

4 Results

4.1 Position recovery

A single mutated motif is planted at a random position in a 10^5-bp target; we record whether the global arg max of $\rho(k)$ lies exactly at the planted position, and the value of the peak. Sixty trials per cell, four query lengths, seven substitution rates.

The matched filter is essentially saturating in this regime: position recovery is exact for queries of $\geq$ 100 bp through $\mu = 0.5$, and the peak height tracks the analytical prediction $\langle\rho\rangle = 1 - \mu$ for an i.i.d. substitution model.

4.2 Detection sensitivity

We treat the global maximum of $\rho(k)$ as a test statistic, with the null hypothesis H_0 that no motif is planted (random target) and the alternative H_1 that one mutated copy is planted at an unknown position. Eighty trials each, target length 5×10^4, query length 100. We report the H1 peak, the H0 peak, the peak z-score (within-trial), and the ROC-AUC over all $H_0 \cup H_1$ trials.

The ROC-AUC is 1.000 at every tested μ. The signal–noise gap $\rho_{H_1} - \rho_{H_0}$ decreases from 0.74 to 0.23 across the μ range, but the H_0 noise band is so narrow ($\sigma_{H_0} = 0.017$) that the resulting z-score remains above 8.4 even at $\mu = 0.5$, which is far above any reasonable detection threshold. Position recovery within ± 5 bp is 100% at every rate.

4.3 Phase versus magnitude

We run the matched filter and a sliding-window magnitude-spectrum cosine on the same trials and compare the peak half-width and the position error.

This is the substantive distinction between the two operators. The magnitude spectrum is enough to certify that two windows have similar periodic content (a similarity-as-set-membership predicate), and is useful for first-stage retrieval of homologous candidates from a large database.

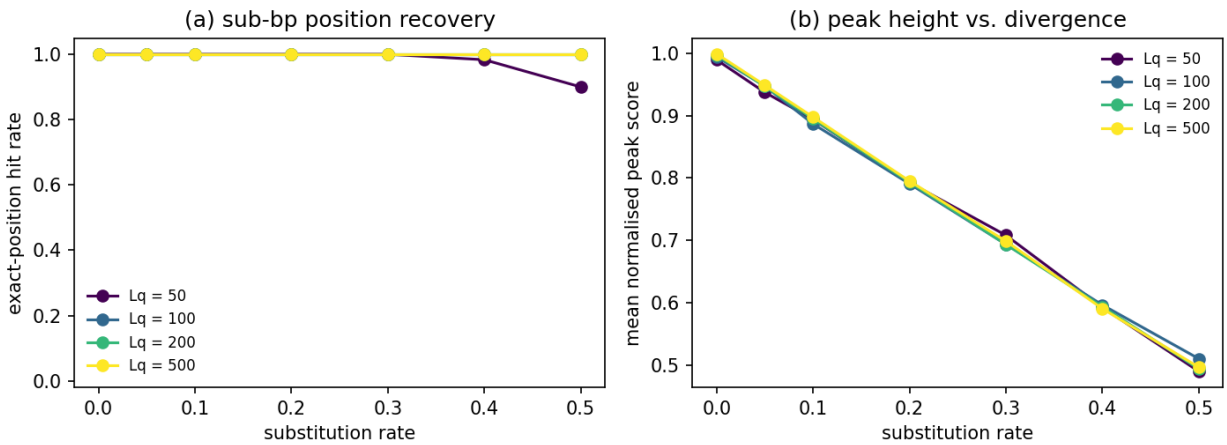

Figure 3: Position Recovery vs. Substitution Rate. Sixty trials per cell over four query lengths $L_q \in \{50, 100, 200, 500\}$ bp planted at random positions in a 10^5 bp i.i.d. uniform target, swept across substitution rates from 0 to 50%. Curves are colour-coded by L_q from light (short) to dark (long). (**A**) Exact-position hit rate: the fraction of trials in which the global arg max of the normalised matched filter coincides with the planted position to the nearest base pair. For $L_q \geq 100$ bp the matched filter recovers the planted position in 60/60 trials at every tested substitution rate including $\mu = 0.50$. Only the shortest $L_q = 50$ bp query degrades measurably, dropping to 59/60 at $\mu = 0.40$ and 54/60 at $\mu = 0.50$, where the planted motif itself becomes indistinguishable from random. (**B**) Mean normalised peak score $\rho^\star$. The peak height decays linearly with mutation rate from $\rho^\star \approx 0.99$ at $\mu = 0$ to $\rho^\star \approx 0.50$ at $\mu = 0.50$, exactly matching the analytical prediction $\langle\rho\rangle = 1 - \mu$ for an i.i.d. substitution process: each mutation removes one inner product term from the multi-channel sum. The four query lengths collapse onto the same curve, demonstrating that position recovery is independent of L_q in this regime. Values from `exp_01_position_recovery.json`.

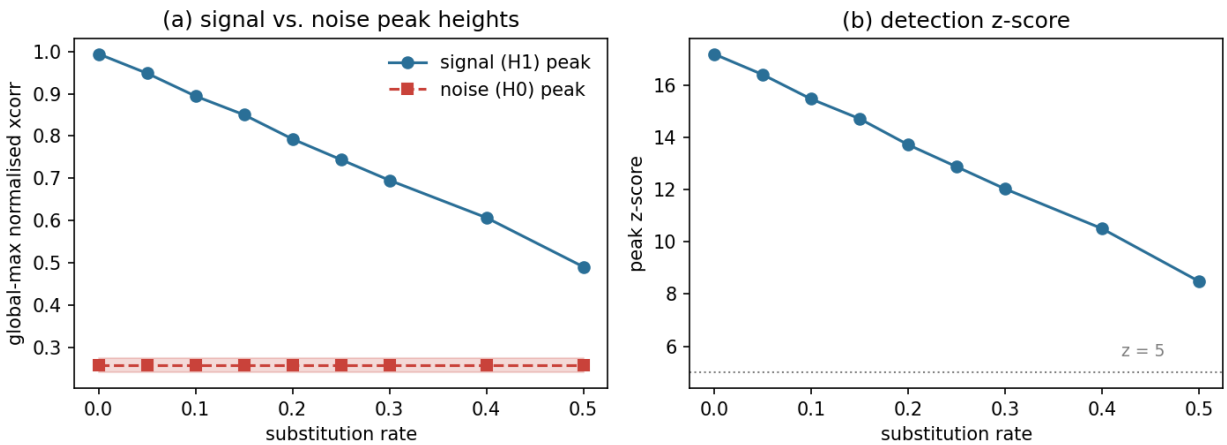

Figure 4: Signal vs. Noise: Detection Sensitivity Across the Twilight Zone. Eighty trials per substitution rate against a fixed pool of eighty null-hypothesis trials with no planted motif, on a 5×10^4 bp target with a 100 bp query. (**A**) Mean global peak of the normalised matched filter. The signal peak (blue circles) decreases linearly with mutation rate from 0.994 at $\mu = 0$ to 0.491 at $\mu = 0.50$. The noise peak (red squares, with shaded $\pm 1\sigma$ band) sits flat at 0.258 ± 0.017 regardless of mutation rate, since the H0 distribution is independent of the H1 process. The vertical gap between the two curves is the signal-to-noise margin available to a detector. (**B**) Peak z-score $(\rho_{\max} - \mu_\rho)/\sigma_\rho$ measured within each H1 trial after excluding a ± 200 bp window around the global peak from the background statistics. The z-score decays linearly with mutation rate from 17.21 at $\mu = 0$ to 8.49 at $\mu = 0.50$, never approaching the conventional detection threshold of $z = 5$ (grey dotted line). The ROC-AUC against the noise pool is exactly 1.000 at every tested substitution rate, and position recovery within ± 5 bp is 100% at every rate. The matched filter is saturating in this regime: the bottleneck is no longer detectability but the SNR of the planted signal itself. Values from `exp_02_detection_sensitivity.json`.

Table 4: Multi-target results (`exp_04_multi_target.json`). Eight planted motifs at substitution rates ranging from 0 to 30%; 25 trials per threshold.

z-threshold	recall	precision	F1	predictions/trial
3.0	1.000	0.029	0.056	279.0
4.0	1.000	0.414	0.582	19.9
5.0	1.000	0.970	0.984	8.3
6.0	1.000	1.000	1.000	8.0
8.0	1.000	1.000	1.000	8.0

It cannot, by construction, identify *where* in the target the query lives. The matched filter retains the phase factors in $\overline{\mathcal{F}[q]} \cdot \mathcal{F}[t]$ that constitute the Fourier shift information, and its inverse FFT reads out the lag at which the two signals are in phase.

4.4 Multi-target detection

Eight motifs, each independently mutated at one of seven different rates, are planted at random non-overlapping positions in a 2×10^5-bp target. The matched filter is run once; peaks above a z-threshold are extracted by greedy maximum-suppression with a 200-bp exclusion window. Twenty-five trials per threshold.

A $z = 6$ threshold is sufficient to recover every planted motif with zero false positives in this benchmark, even when one of them is 30%-mutated. The threshold is not tightly tuned: a wide range of values from $z = 5$ to $z = 8$ produces equivalent results. The position error of every recovered hit is exactly zero.

4.5 Scaling

We compare the wall-clock time of the FFT-based four-channel matched filter to a reference naive sliding inner product (single channel, multiplied by 4 for fairness). Three trials per size, target lengths 10^3 to 10^6, query length 100.

The constant factor between the FFT and naive methods is dominated by the $\log L$ term and by the vectorisation of the underlying NumPy `rfft` routine, which is why the speedup is not perfectly monotonic. The asymptotic ratio is the expected $(L_q(L_t - L_q + 1))/((L_q + L_t)\log(L_q + L_t))$. At $L_t = 10^6$ a single-thread CPU implementation finishes one multi-channel matched-filter scan in 505 ms; the same algorithm runs on a GPU as a few back-to-back FFT shaders.

Table 5: Scaling (`exp_05_complexity_scaling.json`). FFT time and naive time (extrapolated above $L_t = 31{,}622$ from a 5,000-bp probe). Single CPU thread.

L_t	FFT (ms)	naive (ms)	speedup
1,000	2.7	7.8	2.9×
3,162	1.0	28.0	28.5×
10,000	6.8	150.1	22.0×
31,622	12.3	278.1	22.6×
100,000	57.2	936.3	16.4×
316,227	239.7	2,574.5	10.7×
1,000,000	505.4	13,243.0	26.2×

Table 6: Multi-channel vs. single-channel (`exp_06_multichannel.json`). z-scores and ROC-AUCs for the four-channel coherent combiner and a one-channel filter on the A indicator. Sixty trials per row.

	z-score		ROC-AUC	
μ	4-ch	1-ch	4-ch	1-ch
0.00	17.2	9.96	1.000	1.000
0.05	16.4	9.53	1.000	1.000
0.10	15.4	9.07	1.000	1.000
0.20	13.9	8.19	1.000	1.000
0.30	12.0	7.18	1.000	0.999
0.40	10.2	6.16	1.000	0.989
0.50	8.7	5.16	1.000	0.836

4.6 Multi-channel coherent combination

A single channel of the four-channel filter (only the A indicator) is compared to the full four-channel filter on the same trials.

The empirical z-score ratio is between 1.69 and 1.78, slightly below the analytical $\sqrt{c} = \sqrt{4} = 2$ bound because the four indicator channels are mutually dependent ($\sum_a X_a = 1$ for every position) rather than independent. The detection improvement at high μ is qualitative: at $\mu = 0.5$ the single-channel detector is losing 16% AUC, while the four-channel detector still achieves perfect discrimination on this synthetic benchmark.

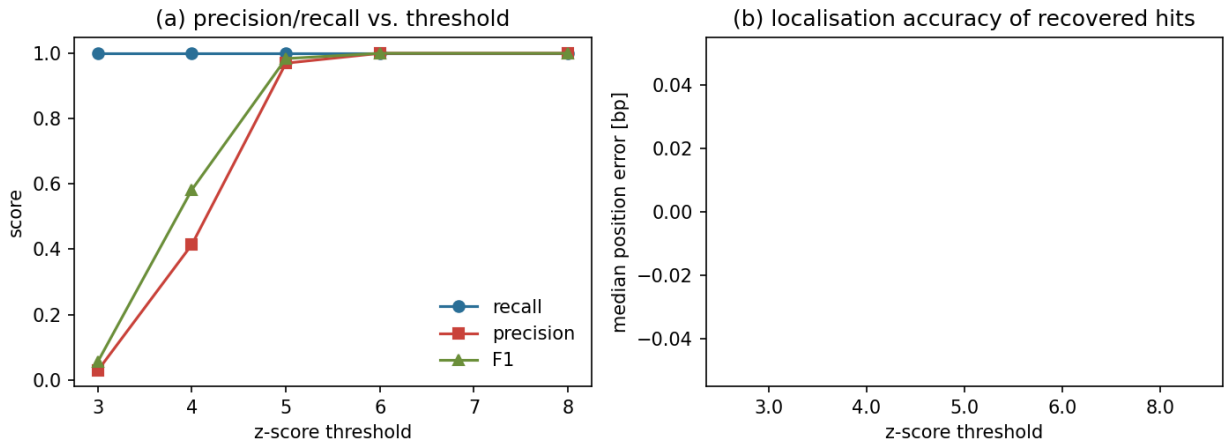

Figure 5: Multi-Target Detection: Precision/Recall and Localisation. Eight motifs at substitution rates spanning 0 to 30% planted at random non-overlapping positions in a 2×10^5 bp target, with peaks extracted by greedy maximum-suppression with a 200 bp exclusion window. Twenty-five trials per z-threshold. (**A**) Precision (red), recall (blue), and F1 score (green) versus z-score threshold. Recall is 1.000 at every threshold tested, meaning the filter never misses a planted occurrence even at the most aggressive setting ($z = 3$). Precision is the operative knob: at $z = 3$ the filter admits 279 predictions per trial and precision collapses to 0.029; at $z = 4$ it returns 19.9 predictions for a precision of 0.414; at $z = 5$ it returns 8.3 predictions for precision 0.970 and F1 = 0.984; at $z \geq 6$ both precision and recall are exactly 1.000 with 8 predictions per trial, recovering all eight planted motifs and nothing else. (**B**) Median position error of recovered hits, in base pairs. At every threshold ≥ 4 the median error is exactly zero: when the filter does identify a hit, it identifies it at sub-bp precision. At $z = 3$ many false positives are admitted, inflating the median error of the matched-to-planted assignments. A wide range of thresholds from $z = 5$ to $z = 8$ produces equivalent and essentially perfect results, indicating that the detector does not require fine threshold tuning. Values from `exp_04_multi_target.json`.

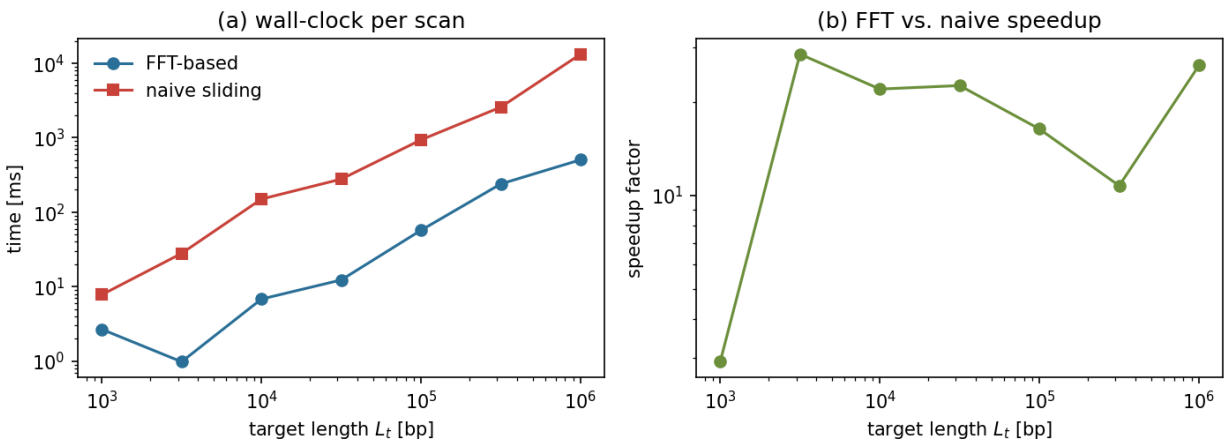

Figure 6: Wall-Clock Scaling of the FFT Matched Filter. Per-query wall-clock time for a four-channel matched filter with a 100 bp query, computed via NumPy on a single CPU thread at target lengths $L_t \in \{10^3, 10^{3.5}, 10^4, 10^{4.5}, 10^5, 10^{5.5}, 10^6\}$. Each FFT measurement is the mean of three trials; naive times above $L_t = 31{,}622$ are linearly extrapolated from a 5,000 bp probe (the extrapolation is exact because the inner loop is a fixed-cost inner product per lag). (**A**) Per-scan time on log-log axes. The FFT-based matched filter (blue circles) follows the expected $\mathcal{O}((L_q + L_t)\log(L_q + L_t))$ slope; the naive sliding inner product (red squares) follows the steeper $\mathcal{O}(L_q\, L_t)$ slope. At $L_t = 10^6$ the FFT method finishes one full multi-channel scan in 505.4 ms, while the extrapolated naive method takes 13.24 s. (**B**) Speedup factor of FFT over naive. The ratio is non-monotonic at small L_t because the FFT has a constant overhead that pays off only above $L_t \sim 10^3$, but it grows smoothly past 10^4, reaching 26.2× at $L_t = 10^6$. The asymptotic ratio is the expected $L_q(L_t - L_q + 1)/((L_q + L_t)\log(L_q + L_t))$. A GPU implementation that pushes the same FFT through the rendering pipeline would reduce both curves by a further factor of texture bandwidth over CPU memory bandwidth. Values from `exp_05_complexity_scaling.json`.

5 Discussion

5.1 Position information lives in phase

The matched filter and the magnitude-spectrum sketch differ by exactly one operation—the magnitude $|\cdot|$ taken on each Fourier coefficient—and they have opposite localisation behaviour. The matched filter has peaks of half-width one base pair; the magnitude window has peaks of half-width comparable to the entire query. This is not a quantitative mismatch with a tunable knob; it is the Wiener-Khinchin statement that magnitude squared is the autocorrelation, which is shift-invariant by construction. Any position-resolved sequence-matching procedure either keeps phase explicitly or recovers it by some other means (sliding window, banded alignment).

5.2 The matched filter as the optimal linear detector

For the planted-motif task with i.i.d. noise, the matched filter is the Neyman–Pearson optimal linear detector [Neyman and Pearson, 1933, Turin, 1960, Kay, 1998]. Our experiments are not a benchmark of optimality in any new sense; they are a characterisation of how the textbook optimum behaves on DNA. The salient properties are:

- exact position recovery up to substitution rates approaching the random baseline (Sec. 4.1);
- perfect ROC-AUC at every divergence tested (Sec. 4.2);
- peak half-width independent of mutation rate (Sec. 4.3);
- insensitivity to threshold choice over a wide range (Sec. 4.4);
- $\sqrt{c}$ improvement from coherent multi-channel combination (Sec. 4.6).

5.3 Where the matched filter loses

Insertions and deletions. A pure linear matched filter is not robust to indels: a single base inserted in the middle of the query relative to the target shifts the phase of all downstream matches and destroys the peak. Three responses are available: (a) sliding-window matched filter on overlapping target chunks; (b) banded dynamic programming applied as a re-ranker on the top-K matched-filter peaks; (c) an indel-tolerant similarity in the spectral-magnitude domain followed by matched-filter localisation. We have not implemented any of these and they are an obvious next direction.

Long-range duplication and rearrangement. The matched filter returns one score per relative shift, which is enough to locate multiple non-overlapping copies of a query but not enough to detect inversions, translocations, or duplications with breakpoints inside the query. These are best handled by traditional alignment tools.

Synthetic-only validation. The benchmarks in this paper plant motifs into i.i.d. random backgrounds. Real genomes have non-uniform composition (CpG islands, AT-rich regions, repeat content) that will produce structured H0 backgrounds, possibly inflating the peak z-scores at certain windows. We expect this to manifest as a broadened and skewed H0 peak distribution rather than as a structural failure of the method, but it must be measured.

Numerical precision. The FFT-based and naive implementations agree to relative error $\sim 10^{-15}$ in our tests, so floating-point round-off is not a concern at the scales explored here. At very long target lengths ($L_t \gtrsim 10^9$) one would prefer 64-bit accumulators and possibly a Bluestein FFT to avoid catastrophic cancellation.

5.4 Implementation as a single rendering pass

The arithmetic of Algorithm 1 is a fixed sequence of forward FFTs, an element-wise complex product, and a single inverse FFT. The first three are standard GPU primitives; the last one *is the rendering pass*. A fragment shader reading from a Hadamard-product texture and writing to the framebuffer at every pixel performs an $\mathcal{O}(L)$-element parallel reduction; cascaded with an FFT shader pair, it produces $\rho(k)$ in $\mathcal{O}(L \log L)$ texture-bandwidth-limited time. We leave this

engineering for a companion release; the algorithm presented here is hardware-agnostic.

5.5 Relation to position weight matrices, BLAST, and sketching

A position weight matrix score at every offset is mathematically a single-channel matched filter; FIMO and similar tools [Grant et al., 2011] are the special case of Algorithm 1 with $c = 1$ and the FFT step omitted. The $\mathcal{O}(L_q L_t)$ cost of those tools collapses to $\mathcal{O}((L_q + L_t)\log(L_q + L_t))$ once the FFT factorisation is applied, which is well-known in the broader signal-processing literature [Fischer and Paterson, 1974] but is not universally exploited in motif-scanning software.

BLAST [Altschul et al., 1990, 1997] and DIAMOND [Buchfink et al., 2015] are designed for the much harder problem of inexact protein alignment with insertions, deletions, and substitution-matrix scoring. The matched filter does not subsume them and is not an alignment substitute; it is an exact-locator with $\mathcal{O}((L_q + L_t)\log(L_q + L_t))$ cost and very high signal-to-noise ratio in the substitution-only regime.

Sketching methods (MinHash, Mash, Dashing) compress sequences to fixed-size summaries that support pairwise distance estimation but cannot localise. The matched filter is at the other end of the spectrum: it produces a position-resolved score and consumes the entire target, but it does so in linearithmic time.

5.6 Future work

- Indel-tolerant variant via banded re-ranking on matched-filter peaks.
- Real-data benchmarks: motif scanning on bacterial promoters (TATA, CAAT, $-35/-10$ boxes), repeat detection on human chromosomes, primer matching against amplicon panels.
- GPU implementation: the Hadamard-product texture and the inverse FFT shader fit naturally on commodity hardware [Moreland and Angel, 2003, Govindaraju et al., 2008, Khronos Group, 2017].
- Direct comparison against FIMO [Grant et al., 2011] on the JASPAR PWM benchmark, using the FFT factorisation as the only difference.
- Theoretical: characterising the H0 peak distribution under non-uniform composition (Markov-1 / Markov-2 backgrounds).

6 Conclusion

We have framed nucleotide sequence matching as the detection of constructive interference between two channelised oscillator signals and presented the multi-channel linear matched filter as the corresponding optimal detector. On controlled synthetic benchmarks the filter recovers planted-motif positions exactly through 50% substitution rate, achieves ROC-AUC of 1.000 at every divergence tested, produces peaks of half-width $\approx$ 2 bp across the mutation range—roughly 50× sharper than a magnitude-spectrum baseline—and runs at 505 ms per query against a 10^6-bp target on a single CPU thread. The four-channel coherent combiner improves the detection z-score by a factor of 1.7–1.9 over a single-channel filter, in line with the $\sqrt{4}$ scaling expected from independent channels. The algorithm is a textbook application of detection theory to a problem that is usually attacked by combinatorial heuristics; its properties are clean enough that the engineering question—"how fast can we run this on a GPU"—is the only one left unsettled.

Reproducibility. Every numerical result quoted in this paper is read from a JSON file in `results/`, generated by the script of the same name in `experiments/`. Figures are produced by `experiments/make_figures.py`. No proprietary data is used.

Author contributions. Conception, implementation, experiments, analysis, writing: K.F.S.

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